MICRO-SLIP PRECONTACT PREDICTION: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

Anonymous authors
Paper under double-blind review

ABSTRACT

We study tactile and force prediction. The motivating bet is: Predict micro-slip risk before stable contact is visually apparent.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For tactile and force prediction, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2024a; 2019; 2022a;b; 2026; 2024b). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a, the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.585	0.415
observed_only_wam	hidden_mode_shift	0.225	0.775
augmentation_as_data_wam	nominal	0.608	0.392
augmentation_as_data_wam	hidden_mode_shift	0.263	0.738
scalar_uncertainty_wam	nominal	0.705	0.295
scalar_uncertainty_wam	hidden_mode_shift	0.338	0.662
proposed_branch_mechanism	nominal	0.777	0.223
proposed_branch_mechanism	hidden_mode_shift	0.537	0.463

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Micro-Slip Precontact Prediction argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

Tactile-driven grasp stability and slip prediction, 2019. https://doi.org/10.3390/robotics8040085.

A soft barometric tactile sensor to simultaneously localize contact and estimate normal force with validation to detect slip in a robotic gripper, 2022a. https://doi.org/10.1109/lra.2022.3205768.

Leveraging distributed contact force measurements for slip detection: a physics-based approach enabled by a data-driven tactile sensor, 2022b. https://doi.org/10.1109/icra46639.2022.9812186.

Tactile sensor-less fingertip contact detection and force estimation for stable grasping with an under-actuated hand, 2024a. https://doi.org/10.1186/s40648-024-00273-3.

Friction coefficient estimation before gross slip for slippage prediction during prosthetic hand grasping, 2024b. https://doi.org/10.1016/j.measurement.2024.114249.

Physical model of electrical contact friction for wear failure analysis and prediction of aerospace slip rings, 2026. https://doi.org/10.1016/j.engfailanal.2026.110526.